

SPI Slave Design

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Overview

- Features

- CPOL=0, CPHA=0
- MSB first
- Back-to-back transfer support
- 8 x 16-bit registers available
- Read data within the same frame
- No system clock
- 24-bit self-defined frame protocol

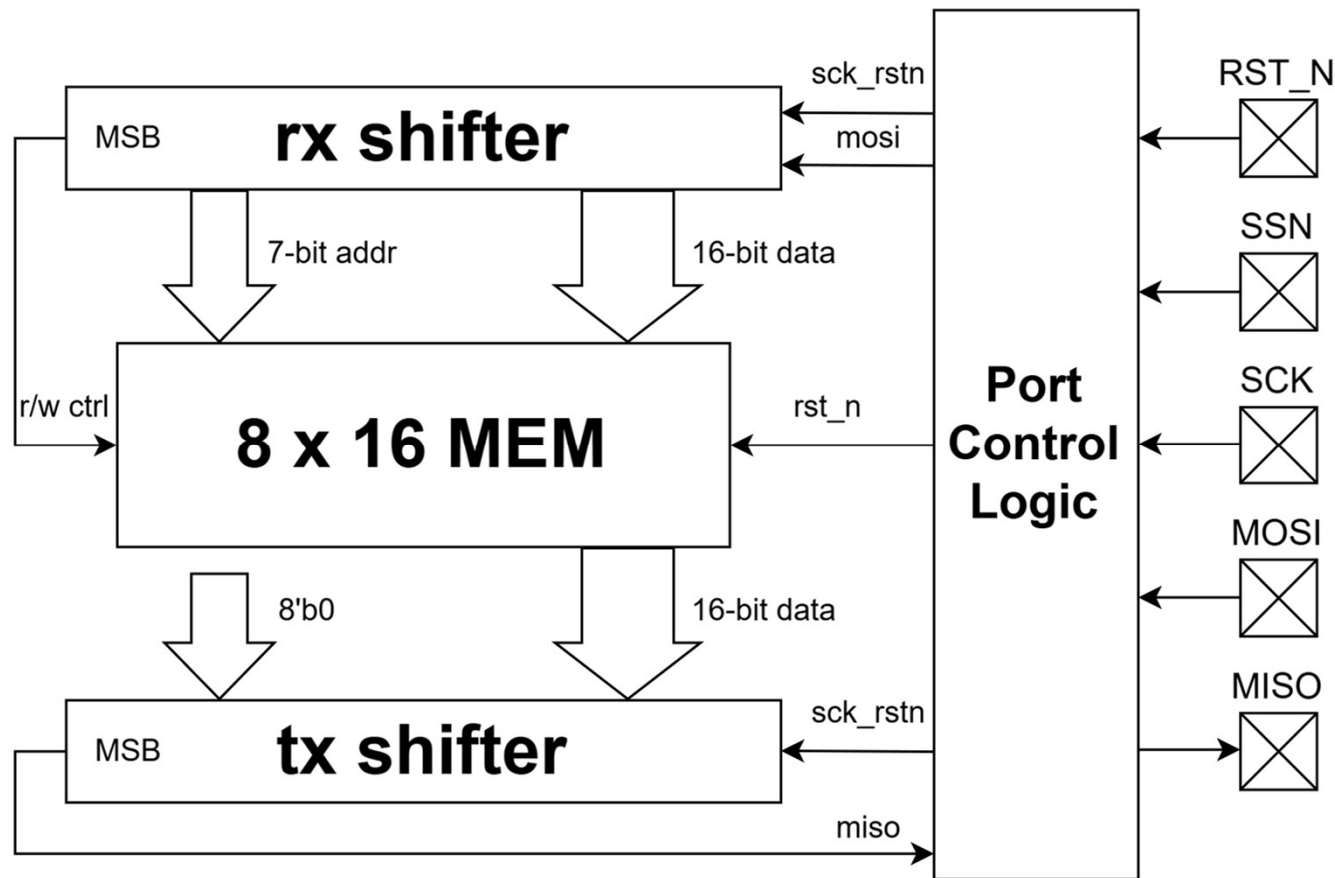
- Frame Structure (MSB first)

- [23] RW 0 = write, 1 = read
- [22:16] ADDR only the low AW bits are decoded, upper bits ignored
- [15:0] DATA

▼ Basic Parameters	
Frame Format	Motorola
Data Size	8 Bits
First Bit	MSB First
▼ Clock Parameters	
Prescaler (for Baud Rate)	8
* Baud Rate	9.0 MBits/s
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge
▼ Advanced Parameters	
CRC Calculation	Disabled
NSS Signal Type	Software

Easy to communicate with MCU using an 8-bit SPI peripheral.
3 Back-to-back transfers per frame with software NSS control.

Front-end Design



SPI Slave Block Diagram

No system clock:

- No CDC problem

```
module spi_slave #(
    parameter integer DW = 16, // Data Width
    parameter integer AW = 3  // Address Width
) (
    input wire rst_n,
    input wire ssn,
    input wire sck,
    input wire mosi,
    output wire miso,
    output wire miso_oen
);

localparam integer CW = 8; // Command Width
localparam integer FW = CW + DW; // Frame Width = 24
localparam integer CNT_W = 5; // Change CNT_W if FW changes
localparam integer REG_N = 1 << AW; // 8 regs, 16 bits each

localparam [CNT_W-1:0] CNT_CMD = CW - 1; // 7, latch command
localparam [CNT_W-1:0] CNT_LOAD = CW; // 8, load tx (neg edge)
localparam [CNT_W-1:0] CNT_LAST = FW - 1; // 23, write mem
```

Design is **parameterized** –
Easy to customize the frame format,
or modify the number of registers.

RTL Simulation

- Simulator: VCS
- Testbench Language: SystemVerilog
- Verification tasks:
 - Reset behavior
 - All zeros/ones transfer
 - Back-to-back transfer
 - Read/Write same addr
 - Read/Write when deselected
 - Randomized multi-frame transfer
 - Functional coverage

```
$finish called from file ".././tb_spi_slave.sv", line 317.  
  
functional coverage = 100.00 %  
checks = 2923 errors = 0 *** PASS ***  
$finish at simulation time 7159920000  
V C S S i m u l a t i o n R e p o r t  
Time: 7159920000 ps  
CPU Time: 0.220 seconds; Data structure size: 0.0Mb  
Mon Aug 24 17:04:14 2026  
functional coverage = 100.00 %  
checks = 2923 errors = 0 *** PASS ***  
[yangh@cryocmoseda tb]$
```

Coverage Report

- Functional Coverage
 - Cover both read and write
 - Cover all 8 regs
 - Cover arbitrary addr_hi
 - Cover data with boundary conditions
 - (Cross Cover) All regs should be both read and written
- Line/Cond/Toggle/Branch Coverage
 - All 100% in VCS coverage report

Summary for Group Instance cg_frame

CATEGORY	EXPECTED	UNCOVERED	COVERED	PERCENT
Variables	18	0	18	100.00
Crosses	16	0	16	100.00

Variables for Group Instance cg_frame

VARIABLE	EXPECTED	UNCOVERED	COVERED	PERCENT	GOAL	WEIGHT	AT LEAST	AUTO BIN MAX	COMMENT
cp_rw	2	0	2	100.00	100	1	1	0	
cp_addr	8	0	8	100.00	100	1	1	0	
cp_arb_addr	2	0	2	100.00	100	1	1	0	
cp_data	6	0	6	100.00	100	1	1	0	

Crosses for Group Instance cg_frame

CROSS	EXPECTED	UNCOVERED	COVERED	PERCENT	GOAL	WEIGHT	AT LEAST	PRINT MISSING	COMMENT
cx_rw_addr	16	0	16	100.00	100	1	1	0	

Cond Coverage for Module : spi_slave

	Total	Covered	Percent
Conditions	26	26	100.00
Logical	18	18	100.00
Non-Logical	0	0	
Event	8	8	100.00

Branch Coverage for Module : spi_slave

	Line No.	Total	Covered	Percent
Branches		13	13	100.00
IF	52	3	3	100.00
IF	70	3	3	100.00
IF	89	3	3	100.00
IF	104	4	4	100.00

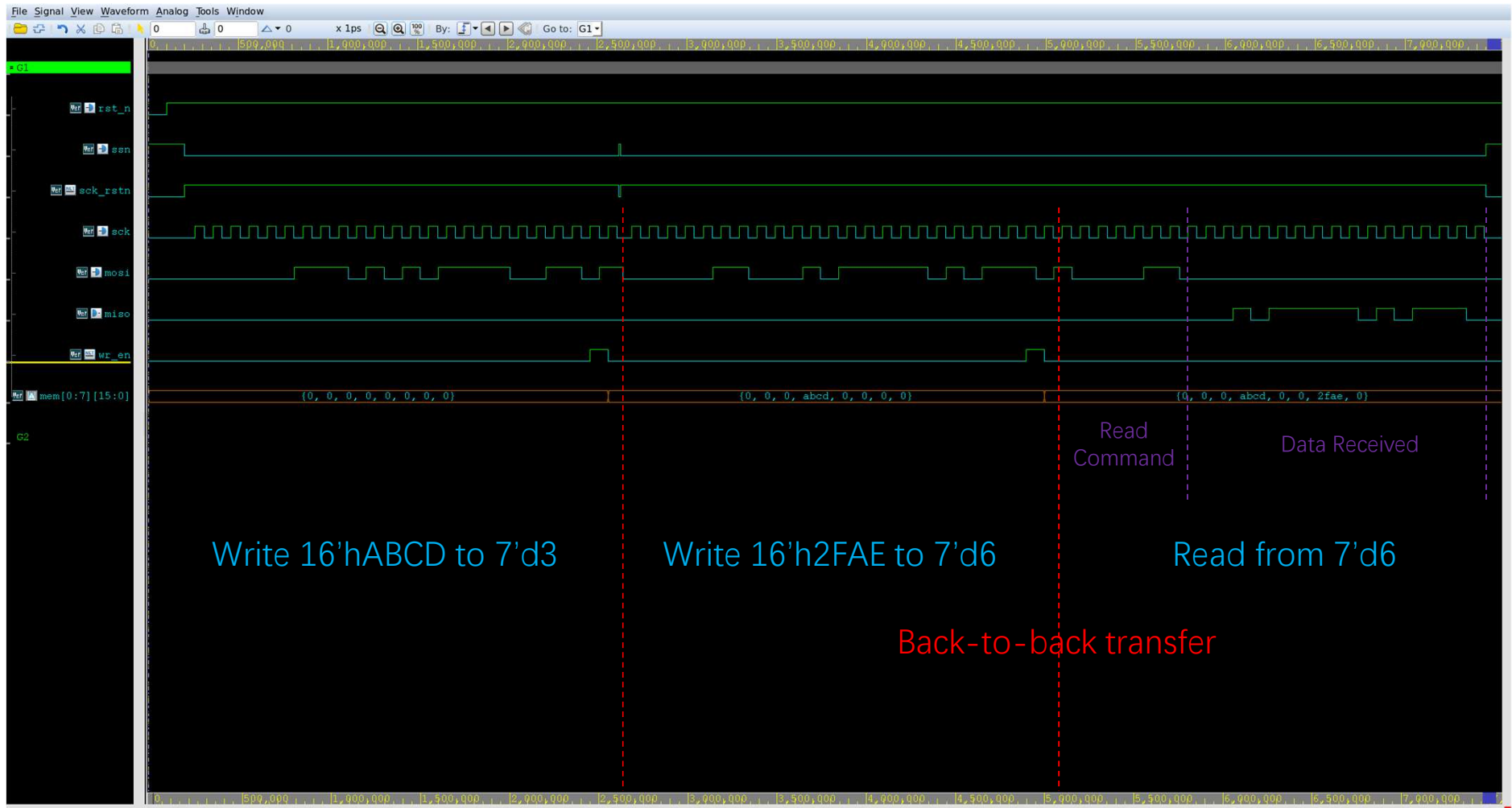
Line Coverage for Module : spi_slave

	Line No.	Total	Covered	Percent
TOTAL		28	28	100.00
CONT_ASSIGN	43	1	1	100.00
ALWAYS	52	5	5	100.00
CONT_ASSIGN	65	1	1	100.00
ALWAYS	70	6	6	100.00
CONT_ASSIGN	83	1	1	100.00
CONT_ASSIGN	84	1	1	100.00
ALWAYS	89	5	5	100.00
CONT_ASSIGN	101	1	1	100.00
ALWAYS	104	5	5	100.00
CONT_ASSIGN	112	1	1	100.00
CONT_ASSIGN	113	1	1	100.00

Toggle Coverage for Module : spi_slave

	Total	Covered	Percent
Totals	17	17	100.00
Total Bits	432	432	100.00
Total Bits 0->1	216	216	100.00
Total Bits 1->0	216	216	100.00
Ports	6	6	100.00
Port Bits	12	12	100.00
Port Bits 0->1	6	6	100.00
Port Bits 1->0	6	6	100.00
Signals	11	11	100.00
Signal Bits	420	420	100.00
Signal Bits 0->1	210	210	100.00
Signal Bits 1->0	210	210	100.00

Waveforms of Transfers



Back-end Design

- Constraints (**Without IO**)

- Clock Frequency: 10MHz
- Clock Uncertainty: $\sim 10\% T$
- Clock Transition: 0.3ns
- **Duty Cycle 60% (60% High, 40% Low)**

- Input/Output delay:

- Max (setup): Assume 50% CLK_LOW for setup
- Min (hold): No requirement as a half-cycle remains before sampling

- Max Fanout: 20

```
// -----  
// Readback and miso.  
// -----  
wire [DW-1:0] rd_data = mem[cmd_addr];  
reg [DW-1:0] tx;  
always @ (negedge sck or negedge sck_rstn) begin // Loads data on falling edge of sck  
    if (!sck_rstn)  
        tx <= {DW{1'b0}};  
    else if (bit_cnt == CNT_LOAD) // On falling edge, cnt already + 1  
        tx <= cmd_rw ? rd_data : {DW{1'b0}};  
    else  
        tx <= {tx[DW-2:0], 1'b0}; // MSB first, fill 0 when idle, so the first byte is always 0x00  
end  
  
assign miso = tx[DW-1];  
assign miso_oen = ssn;
```

Data is transmitted at negedge,
but sampled at posedge.

← Critical path for output delay (setup).

DC & FM

- DC Summary
 - No Violation
 - Area: 12014.33 μm^2
- FM Summary
 - No LAT
 - Verification Succeeded

```
***** Verification Results *****
Verification SUCCEEDED
ATTENTION: synopsys auto setup mode was enabled.
See Synopsys Auto Setup Summary for details.

Reference design: r:/WORK/spi_slave
Implementation design: i:/WORK/spi_slave
170 Passing compare points

Matched Compare Points  BBPin  Loop  BBNet  Cut  Port  DFF  LAT  TOTAL
-----
Passing (equivalent)    0      0      0      0      2    168    0    170
Failing (not equivalent) 0      0      0      0      0      0      0      0
*****

1
#
# Report
#
if { $PASS } {
  puts "\n\n=====
  puts " FORMALITY: VERIFICATION SUCCEEDED"
  puts "=====
} else {
  puts "\n\n=====
  puts " FORMALITY: VERIFICATION FAILED -- see reports"
  puts "=====
}

redirect -tee $RPT_DIR/fm_failing.rpt { report_failing_points }
redirect -tee $RPT_DIR/fm_aborted.rpt { report_aborted_points }
redirect $RPT_DIR/fm_analyze.rpt { analyze_points -all }

save_session -replace $RPT_DIR/fm_fail.fss
}

*****
FORMALITY: VERIFICATION SUCCEEDED
*****

exit
```

```
Timing Path Group 'sck'
-----
Levels of Logic:          0.00
Critical Path Length:     1.48
Critical Path Slack:      17.52
Critical Path Clk Period: 100.00
Total Negative Slack:     0.00
No. of Violating Paths:   0.00
Worst Hold Violation:     0.00
Total Hold Violation:     0.00
No. of Hold Violations:   0.00
-----
```

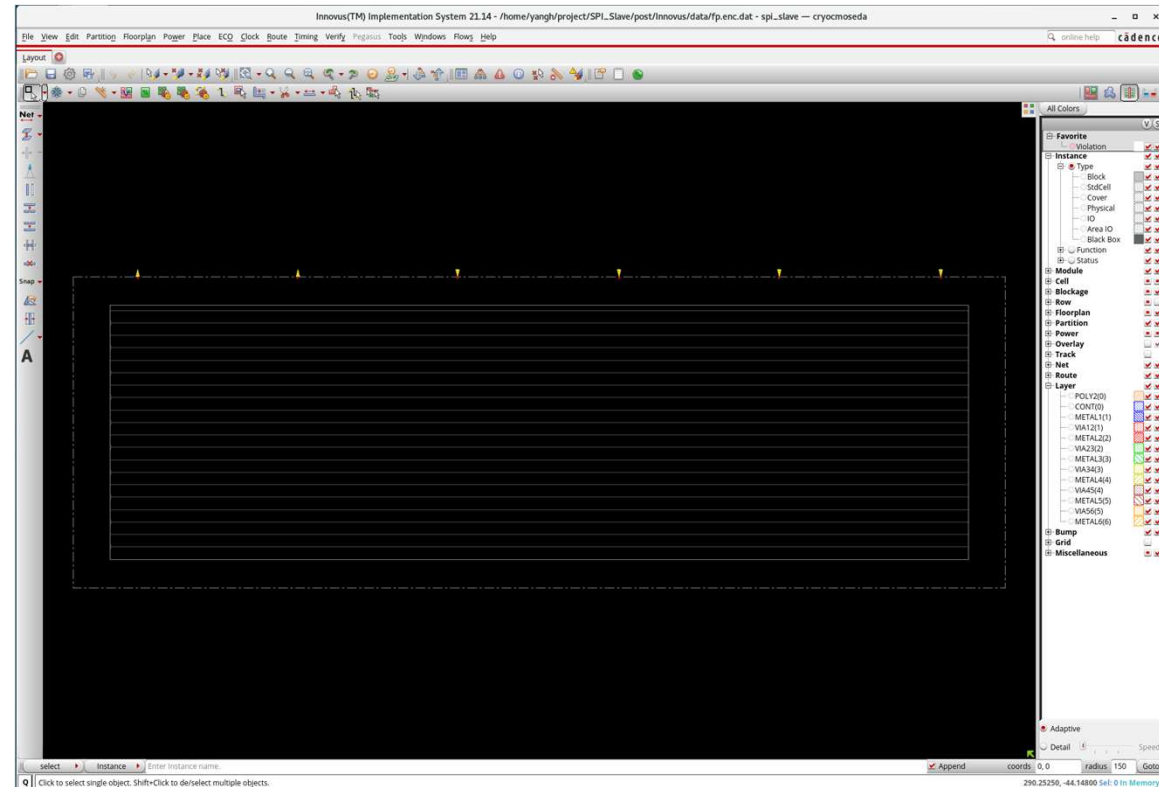
```
Cell Count
-----
Hierarchical Cell Count:  0
Hierarchical Port Count:  0
Leaf Cell Count:          421
Buf/Inv Cell Count:       44
Buf Cell Count:           7
Inv Cell Count:           37
CT Buf/Inv Cell Count:    0
Combinational Cell Count: 253
Sequential Cell Count:    168
Macro Count:              0
-----
```

```
Area
-----
Combinational Area:      3165.478352
Noncombinational Area:   8848.851097
Buf/Inv Area:            305.132795
Total Buffer Area:       61.47
Total Inverter Area:     243.67
Macro/Black Box Area:    0.000000
Net Area:                0.000000
-----
Cell Area:               12014.329449
Design Area:             12014.329449
```

```
Design Rules
-----
Total Number of Nets:    425
Nets With Violations:    0
Max Trans Violations:    0
Max Cap Violations:      0
-----
```

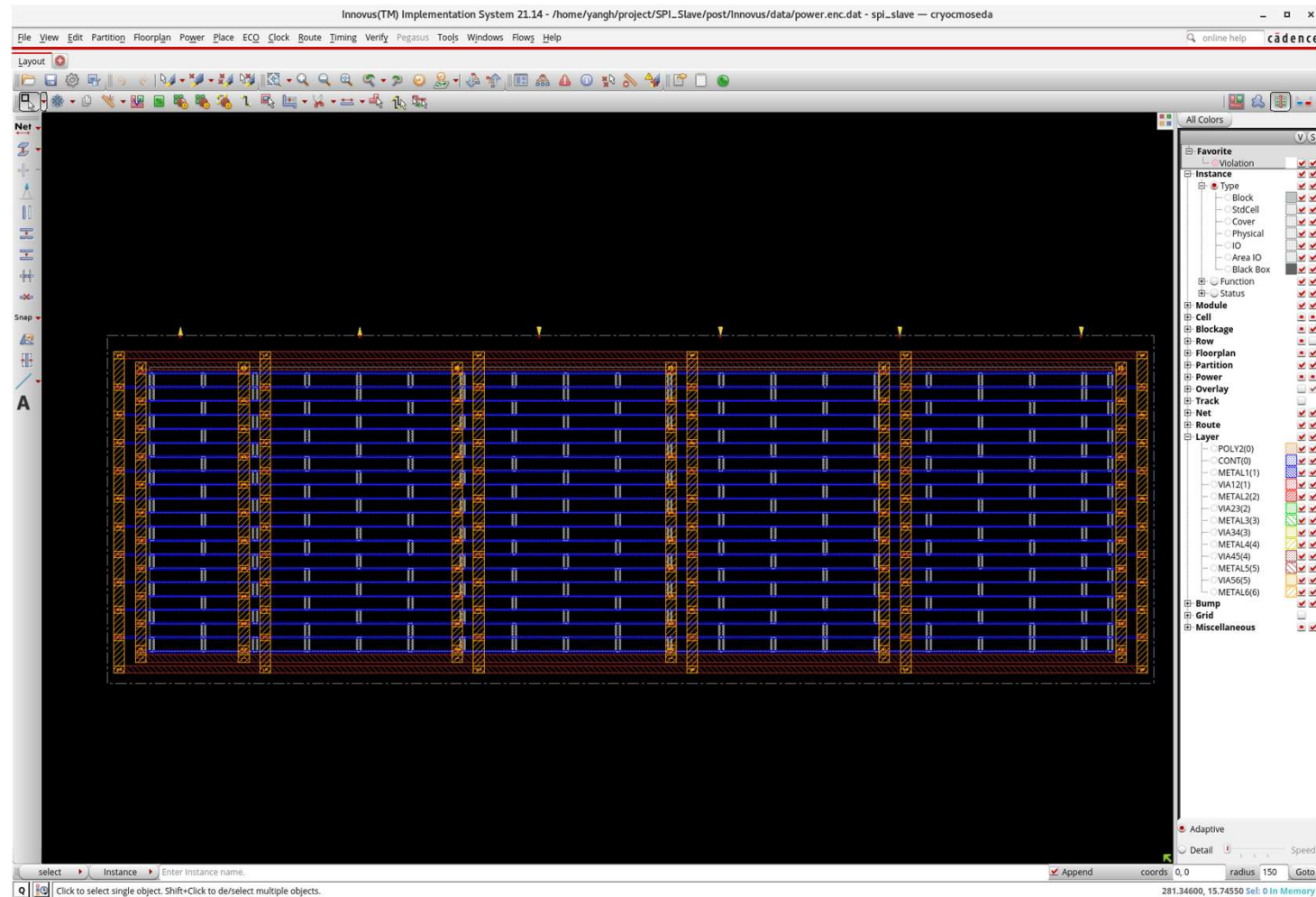

Innovus

- Process: TSMC 0.18um
- LIB: tcb018gbwp7t
 - WC: SS/125°C/1.62V for setup
 - BC: FF/ 0°C / 1.98V for hold
- FloorPlan
 - Die: 294um * 98um
 - $DIE_W = 294 = \text{core } 270 + 2 \times 12 \text{ margin}$
 - $DIE_H = 98 = \text{core } 80 + 2 \times 9 \text{ margin}$
 - Pins: {miso miso_oen mosi sck ssn rst_n}



Innovus

- Power
 - Power Ring
 - Metal 6 Vertical
 - Metal 5 Horizontal
 - Power Stripe
 - Metal 6 Vertical
 - 4 sets
 - Power Rail
 - Metal 1 Horizontal



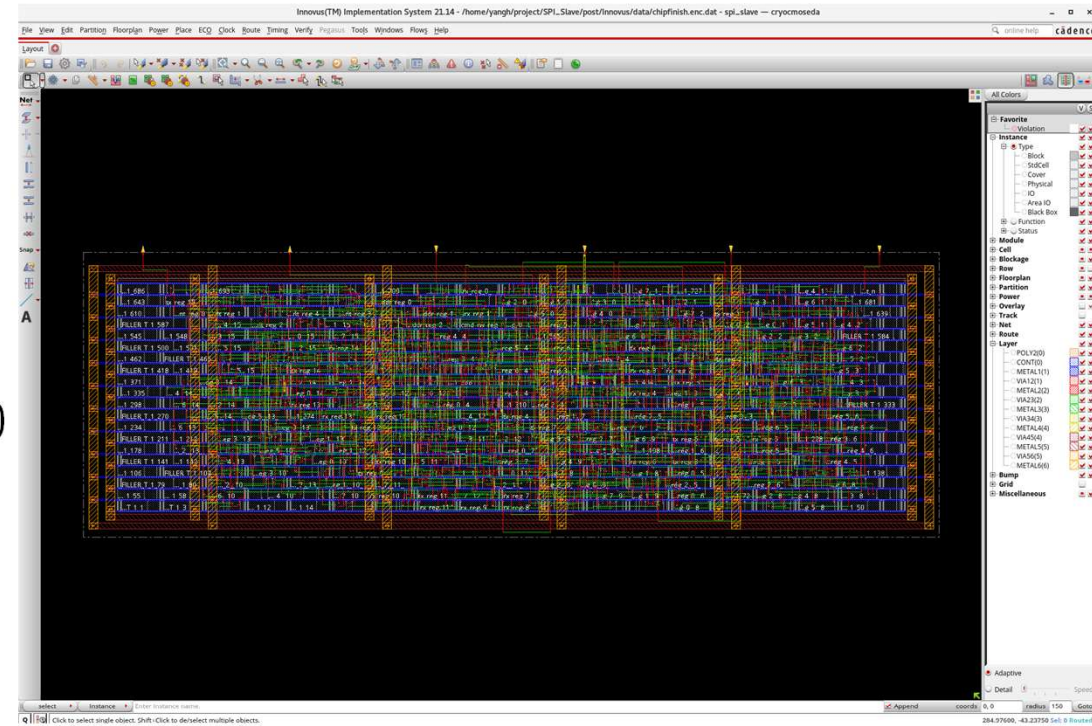
Innovus

- Summary
 - Density: 67.928%
 - 100.000% with Fillers
 - Total number of glitch violations: 0

timeDesign Summary						
Setup mode	all	reg2reg	in2reg	reg2out	in2out	default
WNS (ns):	17.771	54.886	18.300	17.771	78.267	0.000
TNS (ns):	0.000	0.000	0.000	0.000	0.000	0.000
Violating Paths:	0	0	0	0	0	0
All Paths:	391	219	178	1	1	0

DRVs	Real		Total
	Nr nets(terms)	Worst Vio	Nr nets(terms)
max_cap	0 (0)	0.000	0 (0)
max_tran	0 (0)	0.000	0 (0)
max_fanout	0 (0)	0	0 (0)
max_length	0 (0)	0	0 (0)

timeDesign Summary						
Hold mode	all	reg2reg	in2reg	reg2out	in2out	default
WNS (ns):	0.075	0.075	0.444	60.188	0.104	0.000
TNS (ns):	0.000	0.000	0.000	0.000	0.000	0.000
Violating Paths:	0	0	0	0	0	0
All Paths:	391	219	178	1	1	0



IO Plan

- Power IO
 - Pre-drivers & Core
 - 1.8V
 - Post-drivers
 - 3.3V
 - We take PVDD2POC as the post-driver power supply
- Power On Control
 - Will explain later

Table 2.1 Pins Information for Power/Ground Cells

Power/Ground Cells	Connected to	Pin to core	Pad text for LVS
PVDD1DGZ / CDG	core cells + pre-drivers	VDD	VDD :
PVDD2 DGZ / CDG PVDD2POC	post-drivers	None	VD33 : or VDDPST :
PVSS1DGZ / CDG	core cells + pre-drivers	VSS	VSS :
PVSS2DGZ / CDG	post-drivers	None	VSSPST :
PVSS3DGZ / CDG	all the transistors	VSS	VSS :

POC Implementation



Warning: It is mandatory to use ONE and Only One PVDD2POC in each digital domain!

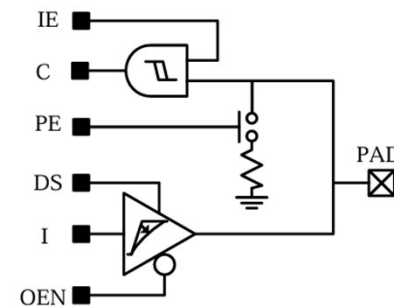
Note: PVDD2POC cell can be either bonded out (as a POC signal generator and the post-driver power supply), or not bonded out (as a filler cell).

IO Plan

- Digital IO
 - Input (PRXW0812SCDG)
 - Pull-up: rst_n, ssn
 - Pull-down: sck, mosi
 - Schmitt Trigger Input
 - For noise immunity
 - Output
 - PRDW0812SCDG
 - With Slew Rate Controlled Output
 - Reduce peak current
 - 8mA output
 - ~7ns edge @20pF, 3.3V

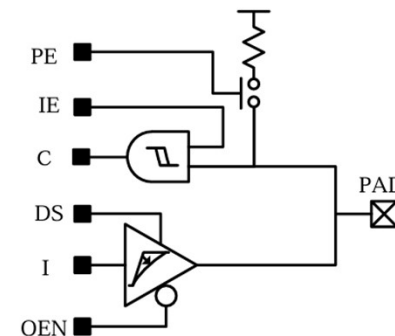
8.23 PRDW0812SCDG

Dual-Driving Regular I/O Cell with Slew Rate Controlled Output, Schmitt Trigger Input, and Enable-Controlled Pull-Down Resistor

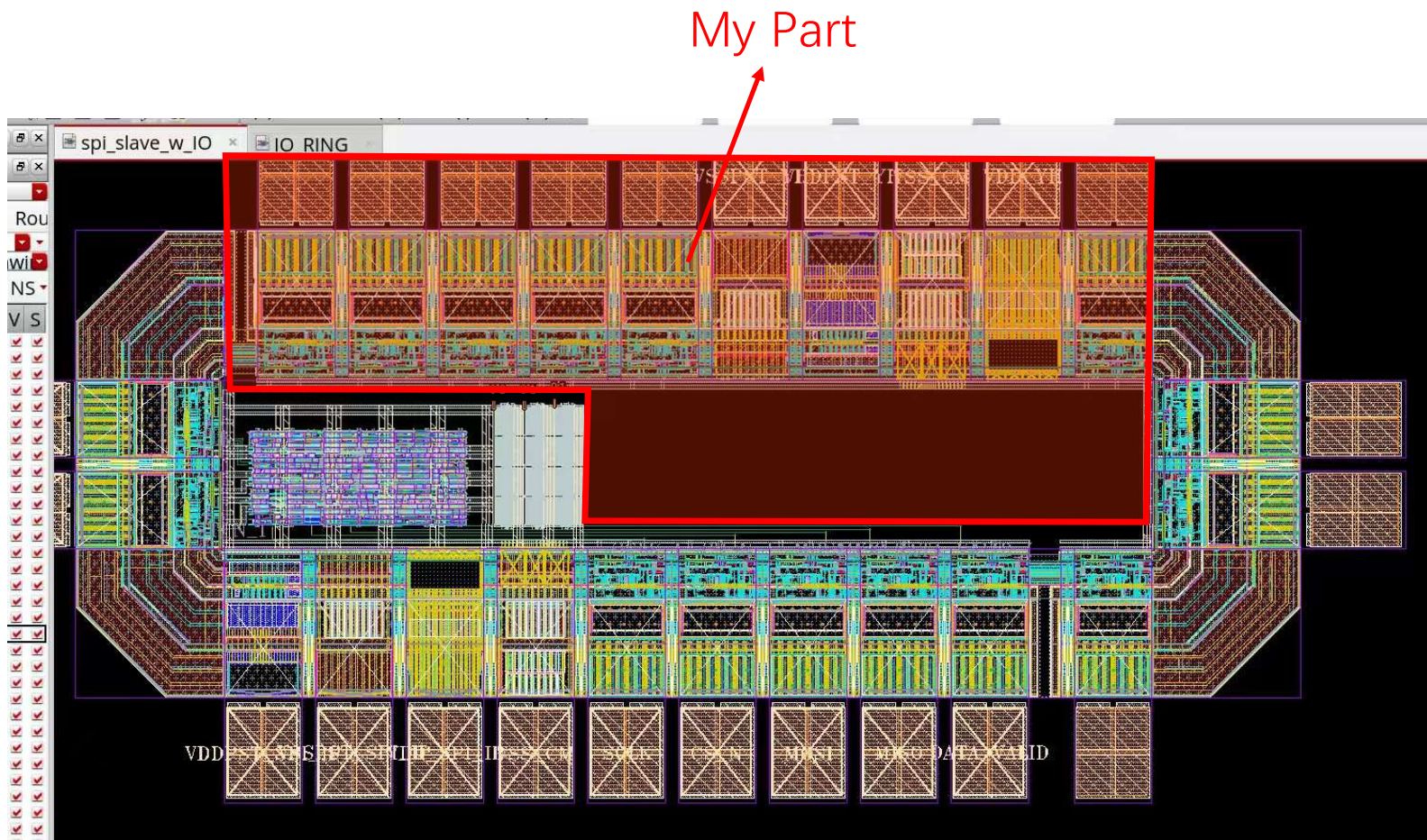


8.31 PRUW0812SCDG

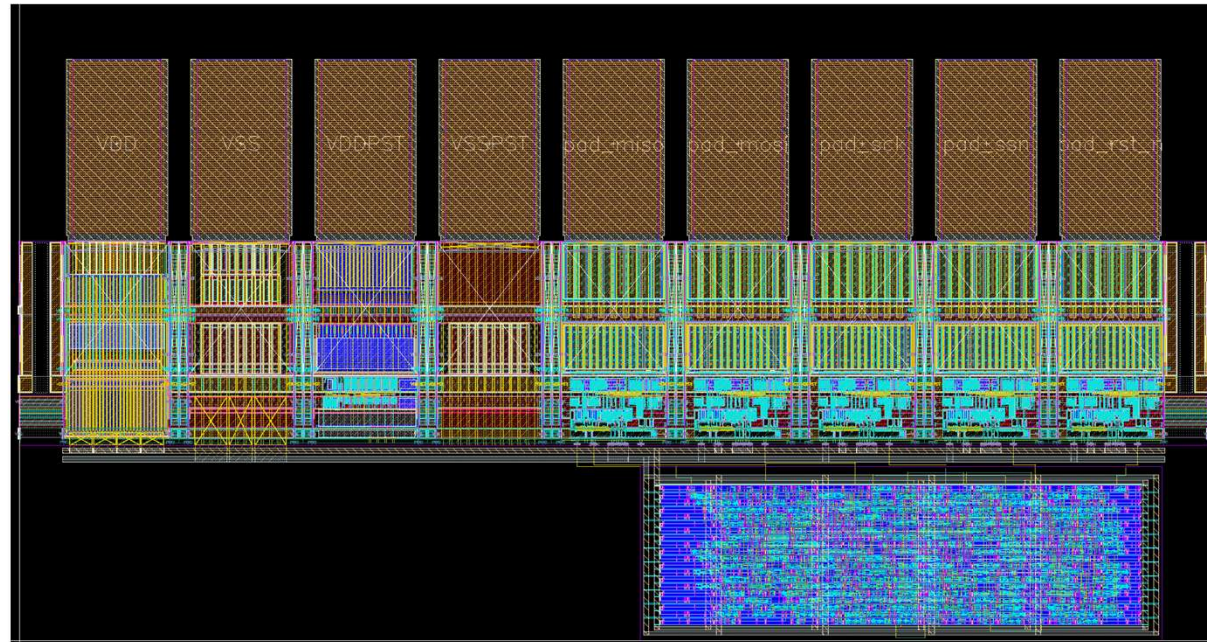
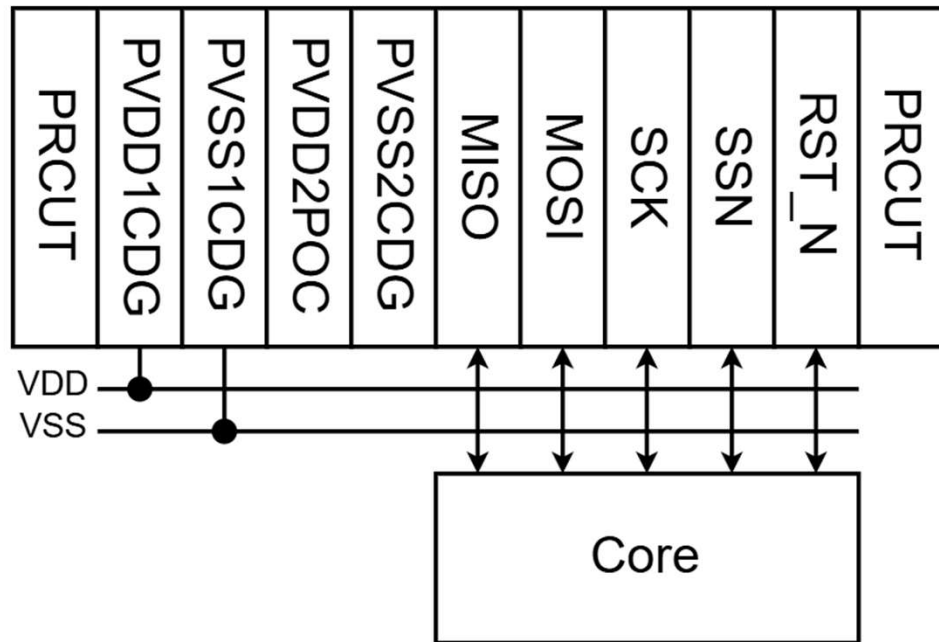
Dual-Driving Regular I/O Cell with Slew Rate Controlled Output, Schmitt Trigger Input, and Enable-Controlled Pull-Up Resistor



Virtuoso Layout



Virtuoso Layout



LVS & DRC

• LVS

- All passed

• DRC

- UTM40K.R.1 { @ Min. density of M6 area $\geq 30\%$ }
 - Dummy fill by others
- PO.R.3 { @ Min poly area coverage $< 14\%$ }
 - Dummy fill by others
- M5.R.1 { @ Min M5 area coverage $< 30\%$ }
 - Dummy fill by others
- ESD.24g & ESD.25g
 - Waived due to rn_tpd018nv_280a.pdf

Check / Cell /	Results	Flat
⊕ × Check UTM40K.R.1	1	1
⊕ × Check PO.R.3	1	1
⊕ × Check M5.R.1	1	1
⊕ × Check ESD.25g	1000	5000
⊕ × Check ESD.24g	30	150

5. KNOWN PROBLEMS AND LIMITATIONS

a. The following DRC rule violations with *T-018-LO-DR-001-C1 v2.8a* could be waived.

- RES.2_PO (I/O cells, PVDD1ANA, PVDD1CDG, PVDD2CDG, PVDD2POC, PXOE1CDG, PXOE2CDG)
- RES.4_PO (I/O cells, PVDD1ANA, PVDD1CDG, PVDD2CDG, PVDD2POC, PXOE1CDG, PXOE2CDG)
- ESD.24g (I/O cells, PXOE1CDG, PXOE2CDG)
- ESD.25g (I/O cells, PXOE1CDG, PXOE2CDG)

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Gate-level Simulation

- Simulator: VCS
- Testbench Language: SystemVerilog
- Same tasks with SDF annotation
- Max/Min all passed

```
Command: /home/yangh/project/SPI_Slave/tb/sim/gls_max/./simv +TEST=all +ntb_random_seed=1 -l run_all_1.log
Chronologic VCS simulator copyright 1991-2021
Contains Synopsys proprietary information.
Compiler version S-2021.09-SP1_Full64; Runtime version S-2021.09-SP1_Full64; Aug 24 21:26 2026
Doing SDF annotation ..... Done
$finish called from file "../../tb_spi_slave.sv", line 317.

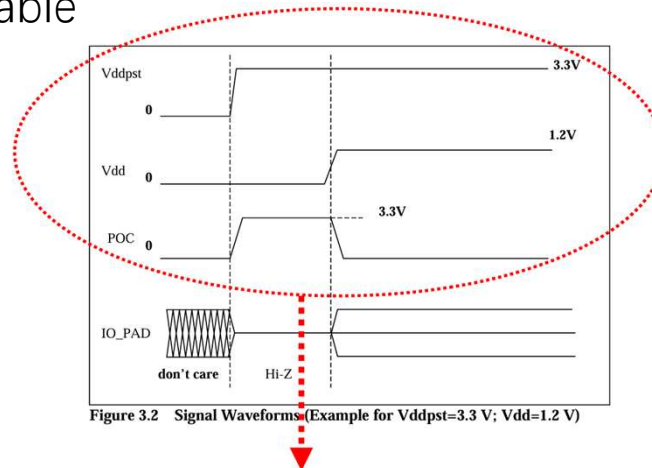
functional coverage = 100.00 %
checks = 2923 errors = 0 *** PASS ***
$finish at simulation time 7159920000
V C S S i m u l a t i o n R e p o r t
Time: 7159920000 ps
CPU Time: 1.810 seconds; Data structure size: 0.3Mb
Mon Aug 24 21:26:40 2026
```

```
Command: /home/yangh/project/SPI_Slave/tb/sim/gls_min/./simv +TEST=all +ntb_random_seed=1 -l run_all_1.log
Chronologic VCS simulator copyright 1991-2021
Contains Synopsys proprietary information.
Compiler version S-2021.09-SP1_Full64; Runtime version S-2021.09-SP1_Full64; Aug 24 21:26 2026
Doing SDF annotation ..... Done
$finish called from file "../../tb_spi_slave.sv", line 317.

functional coverage = 100.00 %
checks = 2923 errors = 0 *** PASS ***
$finish at simulation time 7159920000
V C S S i m u l a t i o n R e p o r t
Time: 7159920000 ps
CPU Time: 1.540 seconds; Data structure size: 0.3Mb
Mon Aug 24 21:26:44 2026
```

Post-layout Simulation

- POC test
- Why POC?
 - The power-up sequence between rails is not guaranteed
 - It is common for the IO supply (3.3 V) to come up before the core supply (1.8 V)
 - POC is required to clamp the IO into a known state (typically high-Z) until the core supply is stable



Note: Details on the relationship among VDD, VDDPST, and POC are addressed on the following page.

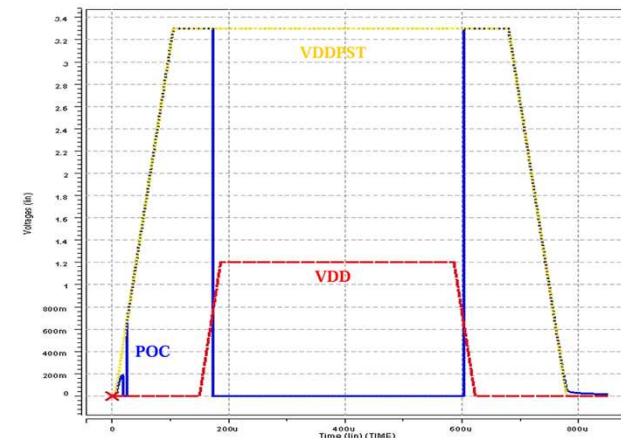
Power On Control Usage

TSMC I/O libraries contain the Power On Control (POC) system to prevent I/O unknown state. An *unknown state* may cause high I/O crowbar current or bus contention when the I/O voltage is powered up before the core voltage.

The PVDD2POC cell features POC circuitry for POC signal generation:

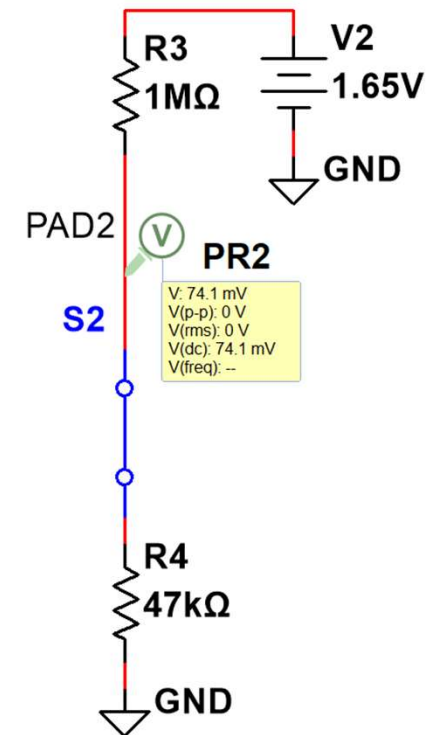
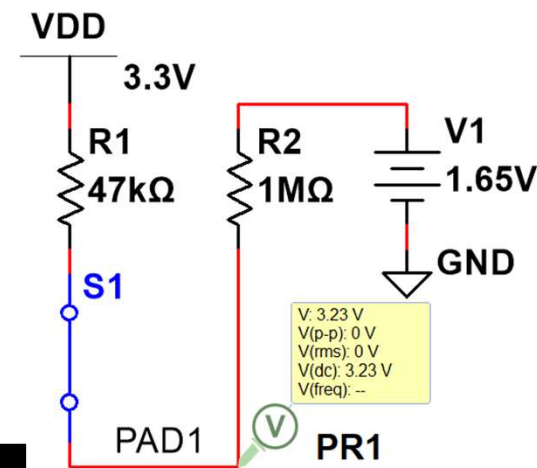
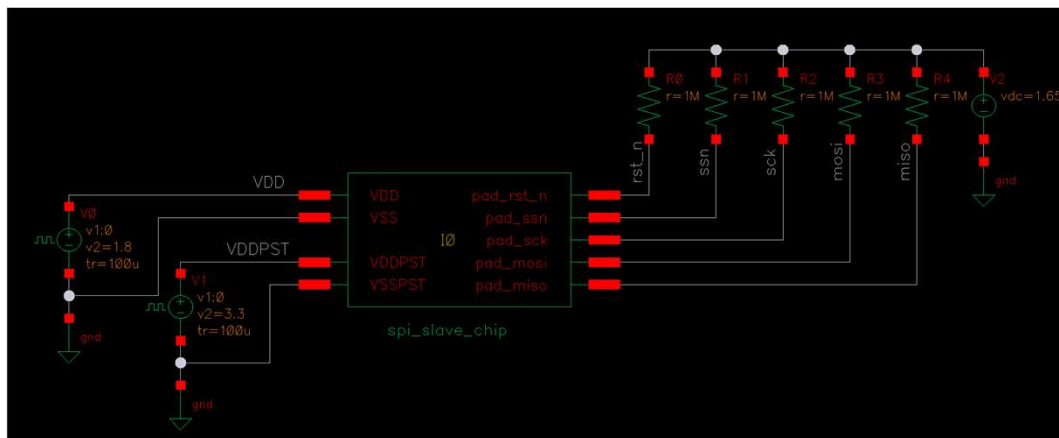
- The **POC circuitry** is located within PVDD2POC cell, which plays the similar role to PVDD2DGZ / CDG cell for I/O-voltage power supply and for ESD protection between post-driver power rail and post-driver ground rail. PVDD2POC detects the “power-on” condition and generates a POC signal to the rest of the I/Os.
- The **POC signal** is distributed to other I/O buffers by cell abutment; no special POC routing is required.

For standard I/O cell, the POC signal can turn off the output NMOS and PMOS buffers when I/O power is on whereas core power remains off. For example, during the power up sequence, the I/O cell is set in the Hi-Z state or the input state.



POC Test

- How to test high-Z:
 - Connect IO to 1.65V with 1M resistor
 - If pad is high-Z, it'll be clamped at 1.65V
 - If pad is controlled by the core, it'll be either near 3.3V / 0V



POC Test @ tt/1.8V/3.3V/27°C

Tue Aug 25 00:13:35 2026

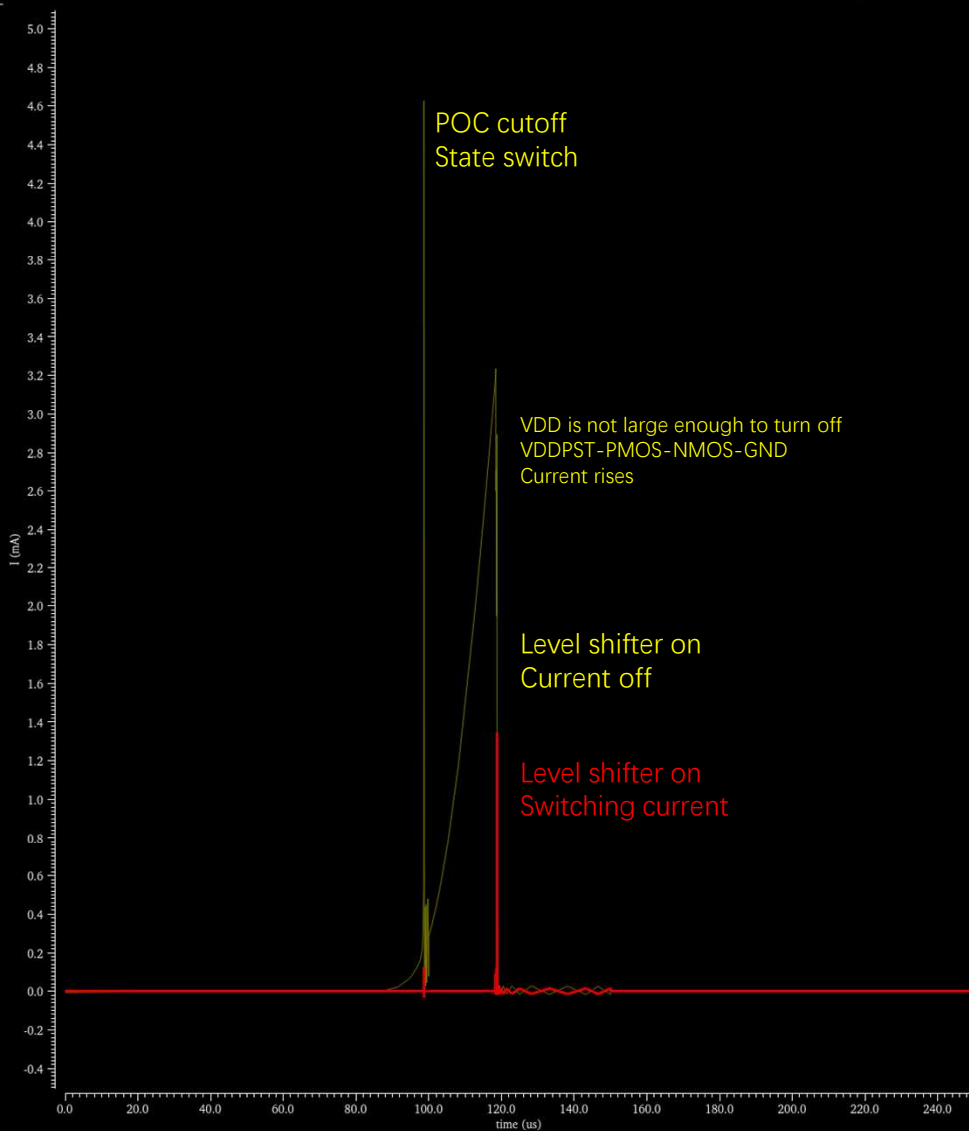
1 Transient Response

Tue Aug 25 00:13:35 2026

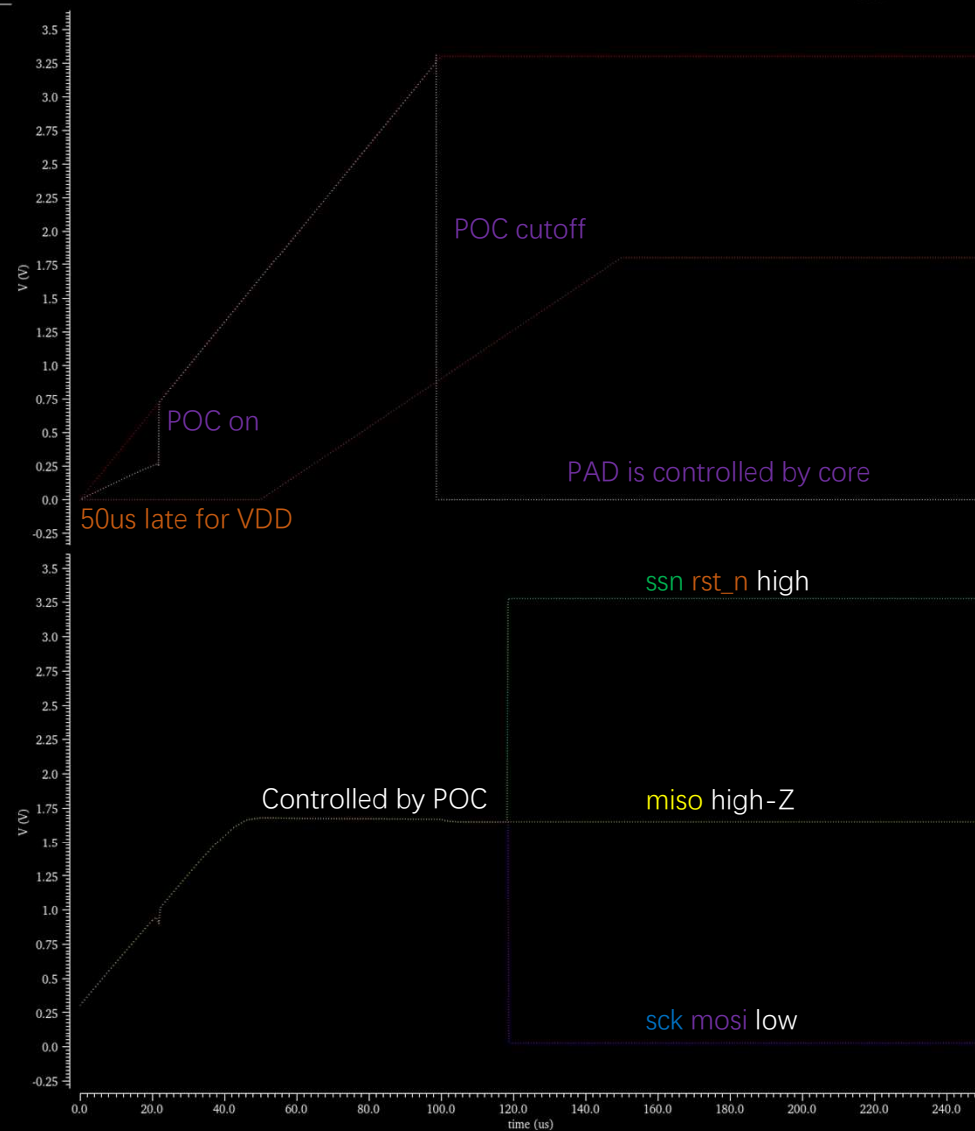
2

lvdd:lvddpst

Name	Vis
lvdd	☒
lvddpst	☒

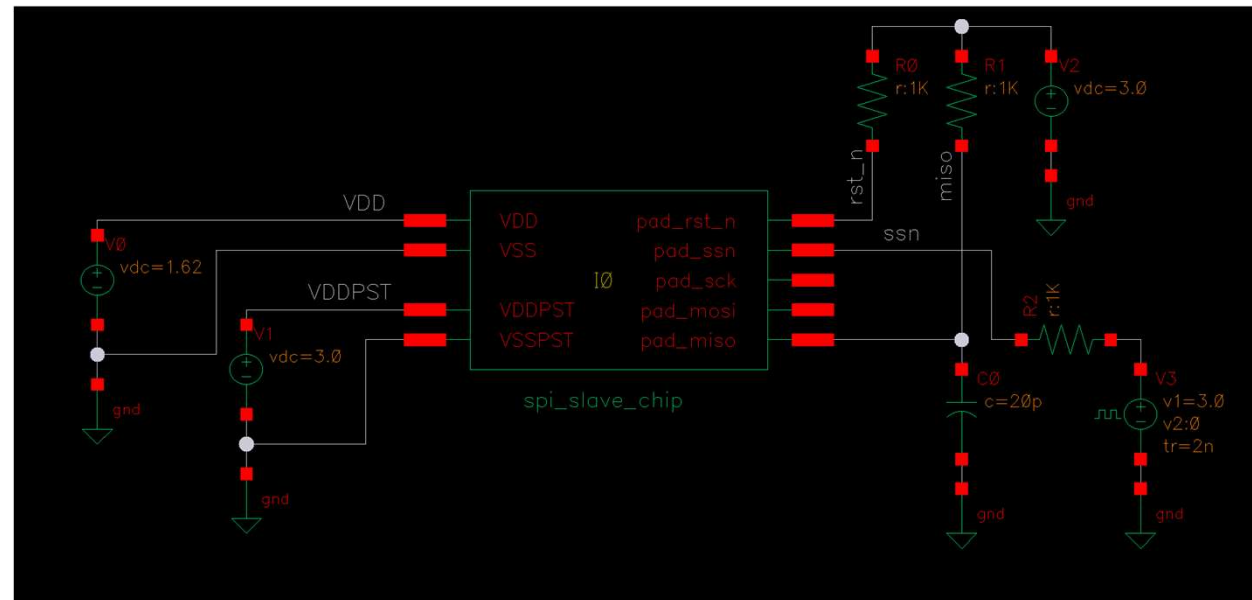


Name	Vis
VDD	☒
VDDPST	☒
/IO/POC	☒

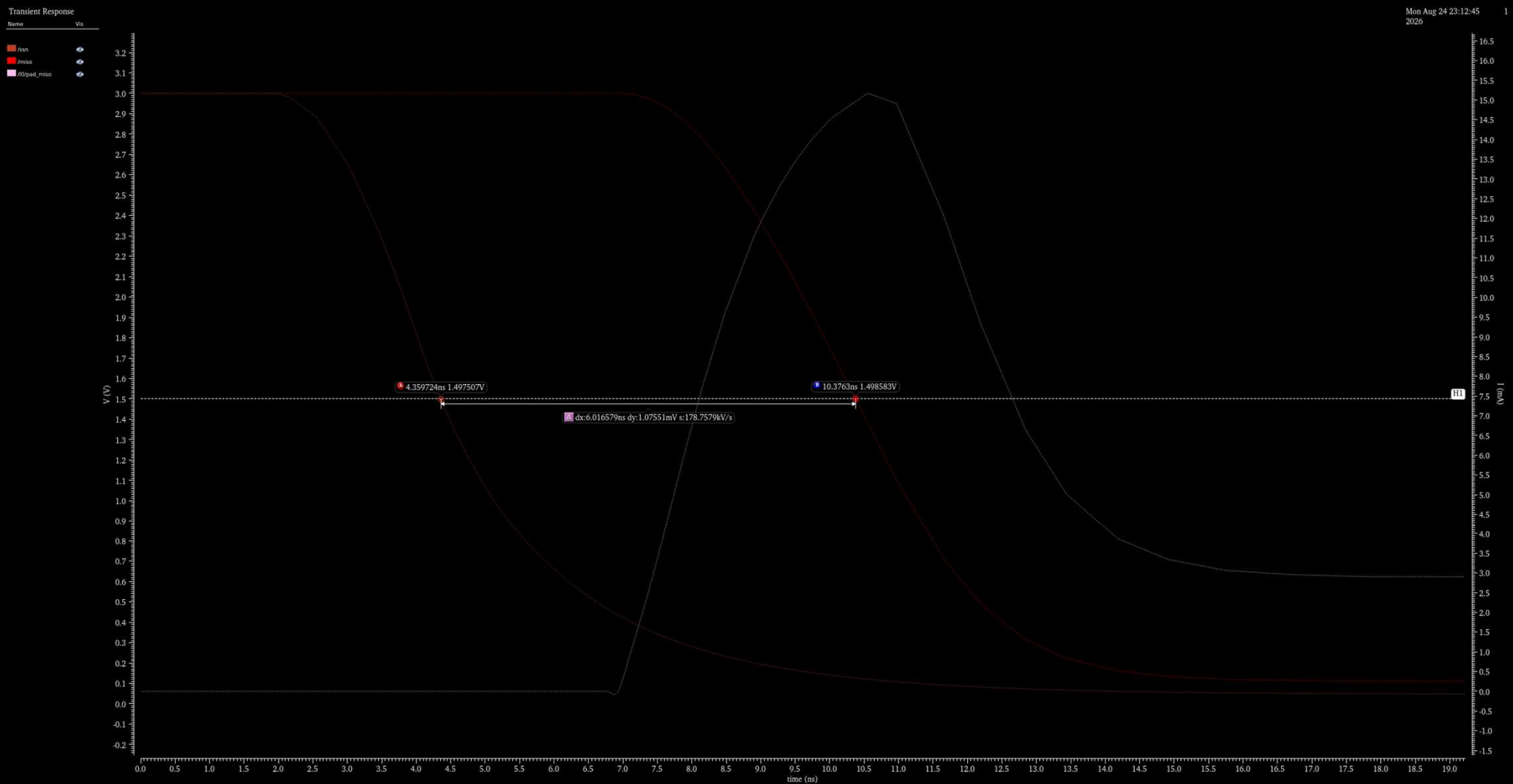


Critical Timings

- t_{EN} : Output enable time
 - Maximum delay from the falling edge of ssn until miso leaves the high-Z state and is actively driven
 - Measured from the 50% point of the ssn waveform to the 50% point of the miso waveform

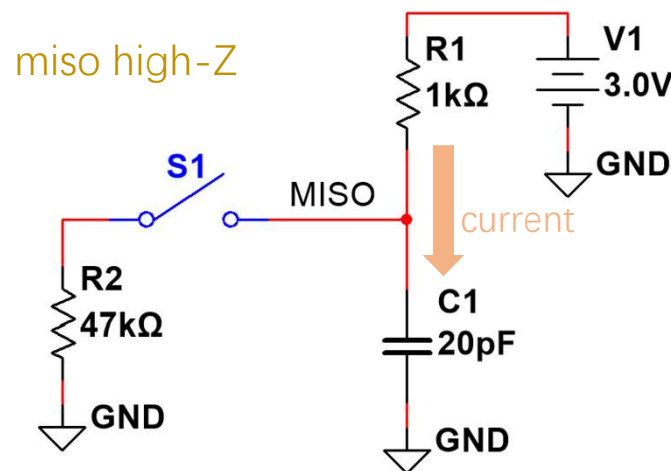
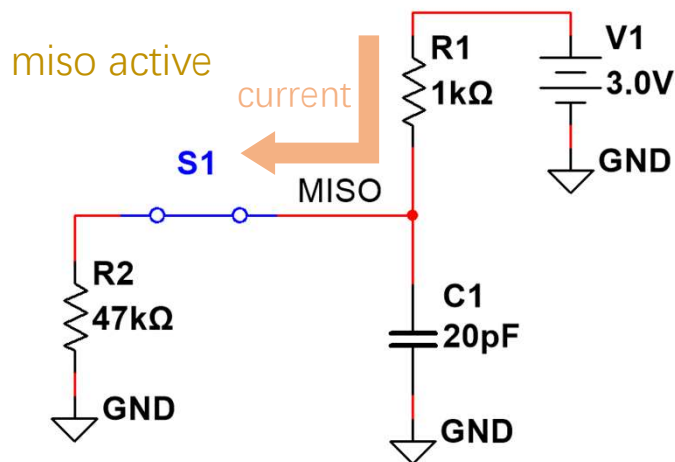


$$t_{EN} = 6.02\text{ns} \quad @ \text{ ss}/1.62\text{V}/3.0\text{V}/125^{\circ}\text{C}, C_L=20\text{pF}$$



Critical Timings

- t_{DIS} : Output disable time
 - Maximum delay from the rising edge of ssn until miso returns to the high-Z state
 - Measured the current from stable to approximately 0
 - Time for miso to return to high-Z voltage is determined by the external R and C_L

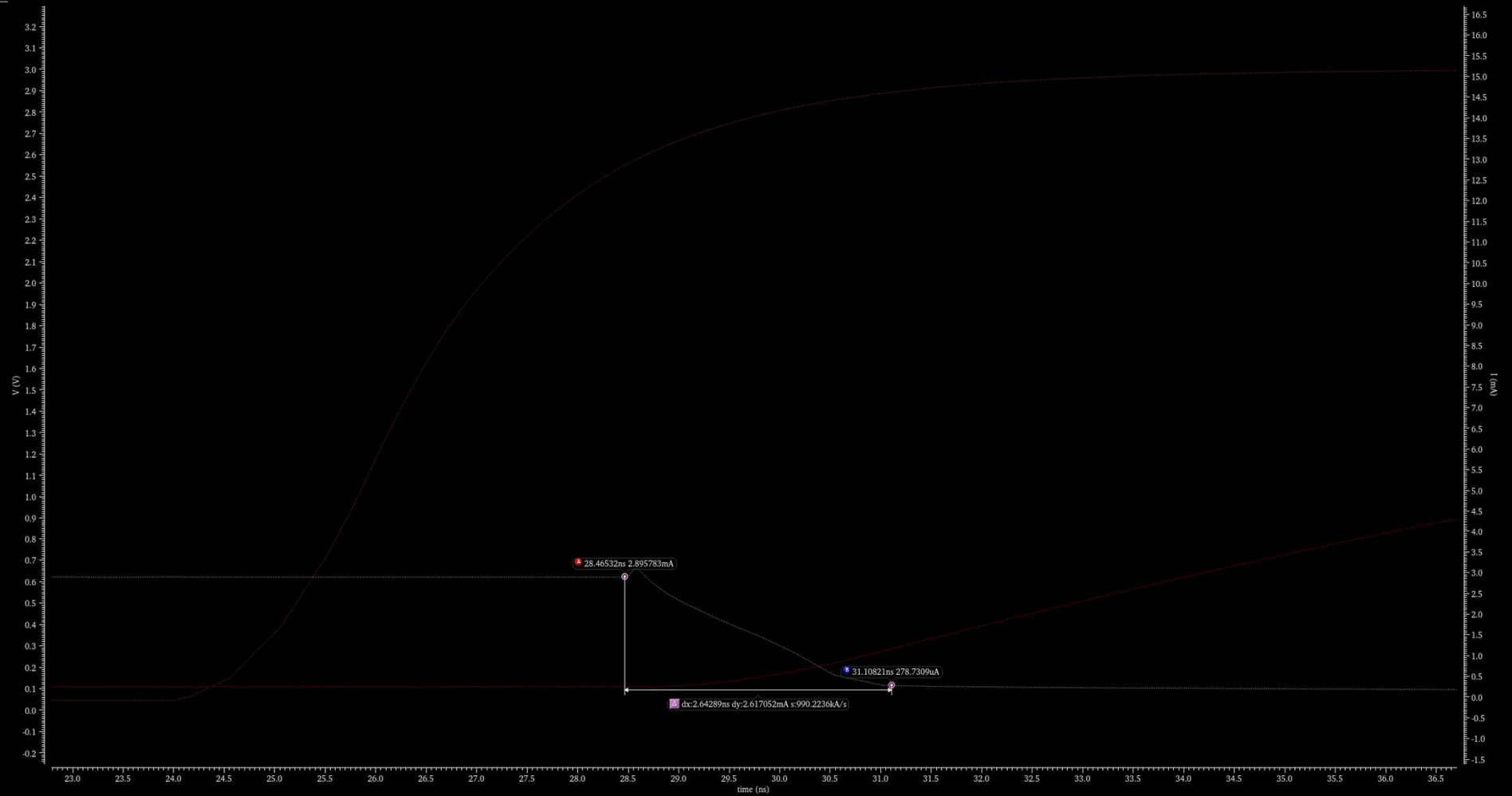


$t_{DIS} = 2.64\text{ns}$ @ ss/1.62V/3.0V/125°C, $C_L=20\text{pF}$

Transient Response

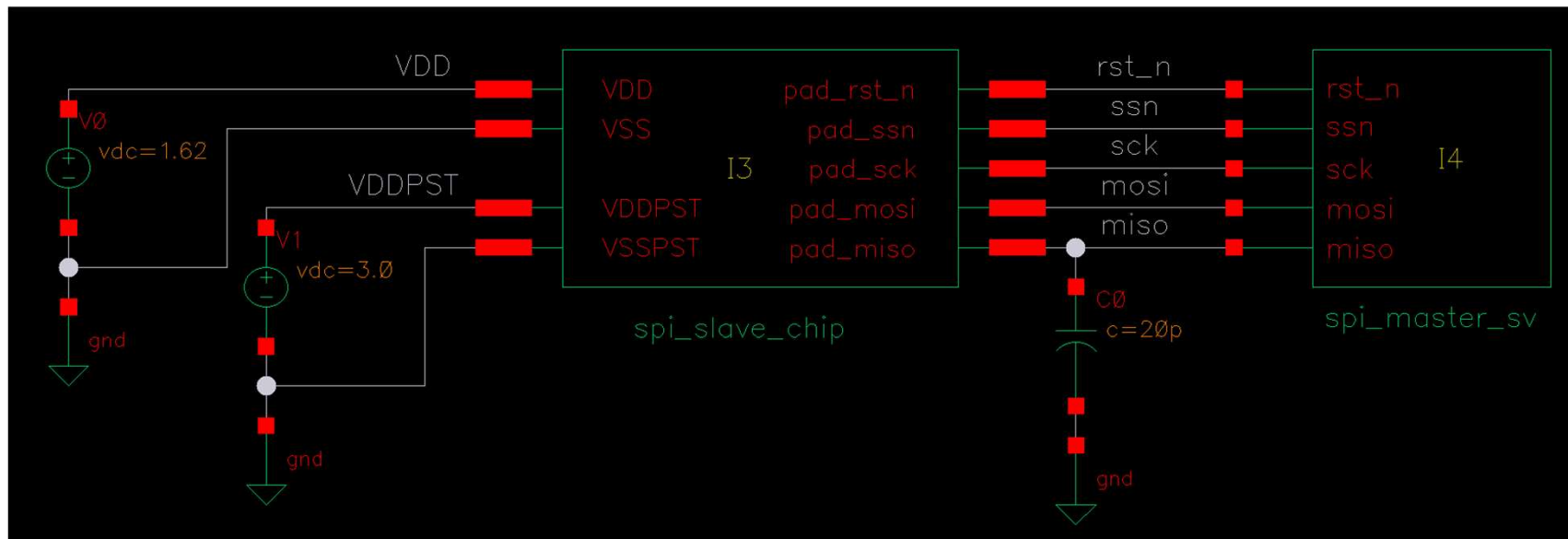
Name
/sim
/ratio
R01pad_miso

Tue Aug 25 09:11:26 2026 1

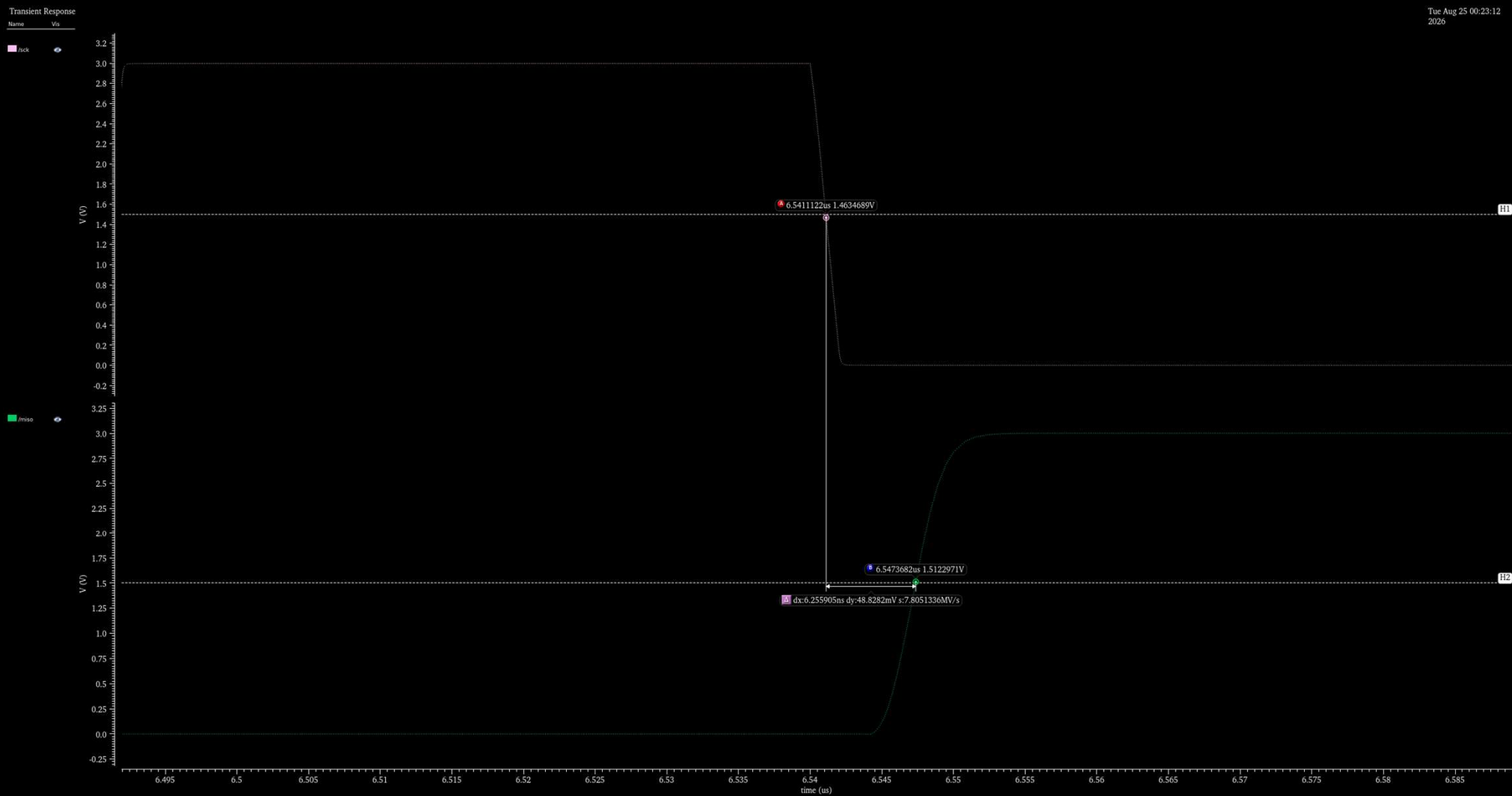


Critical Timings

- t_v : Data output valid time
 - Maximum delay from the falling edge of sck to miso becoming valid
 - Measured from the 50% point of the sck waveform to the 50% point of the miso waveform

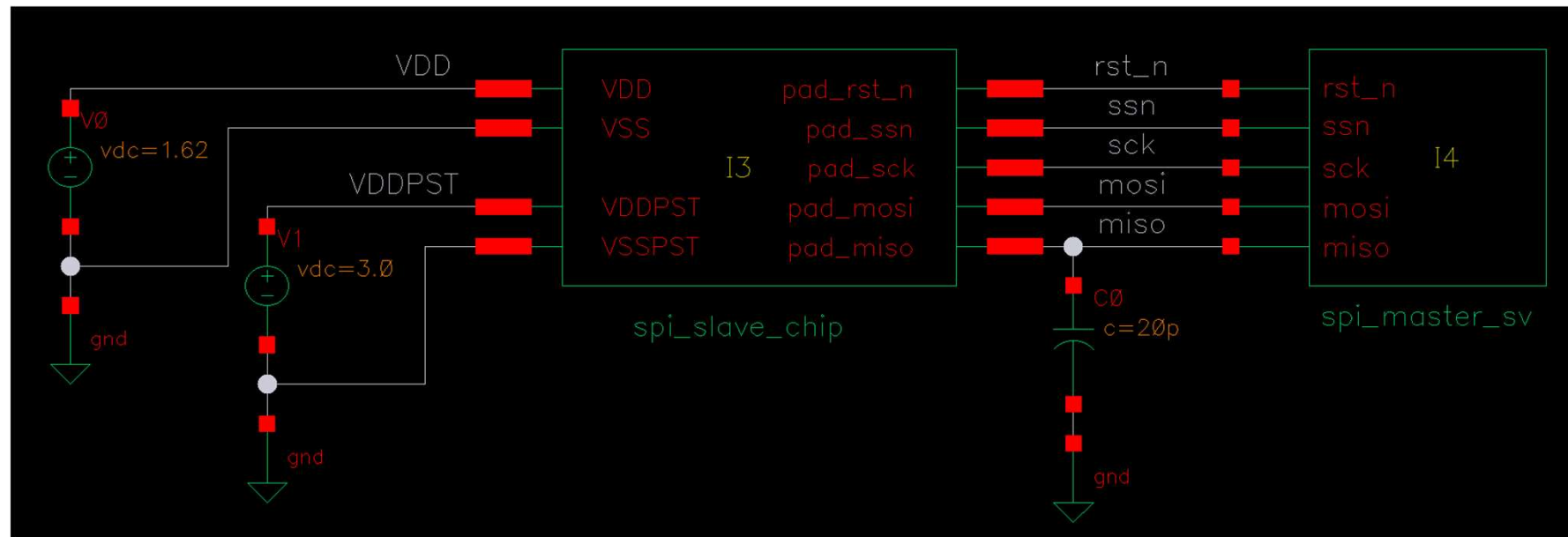


$t_v = 6.26\text{ns}$ @ ss/1.62V/3.0V/125°C, $C_L=20\text{pF}$

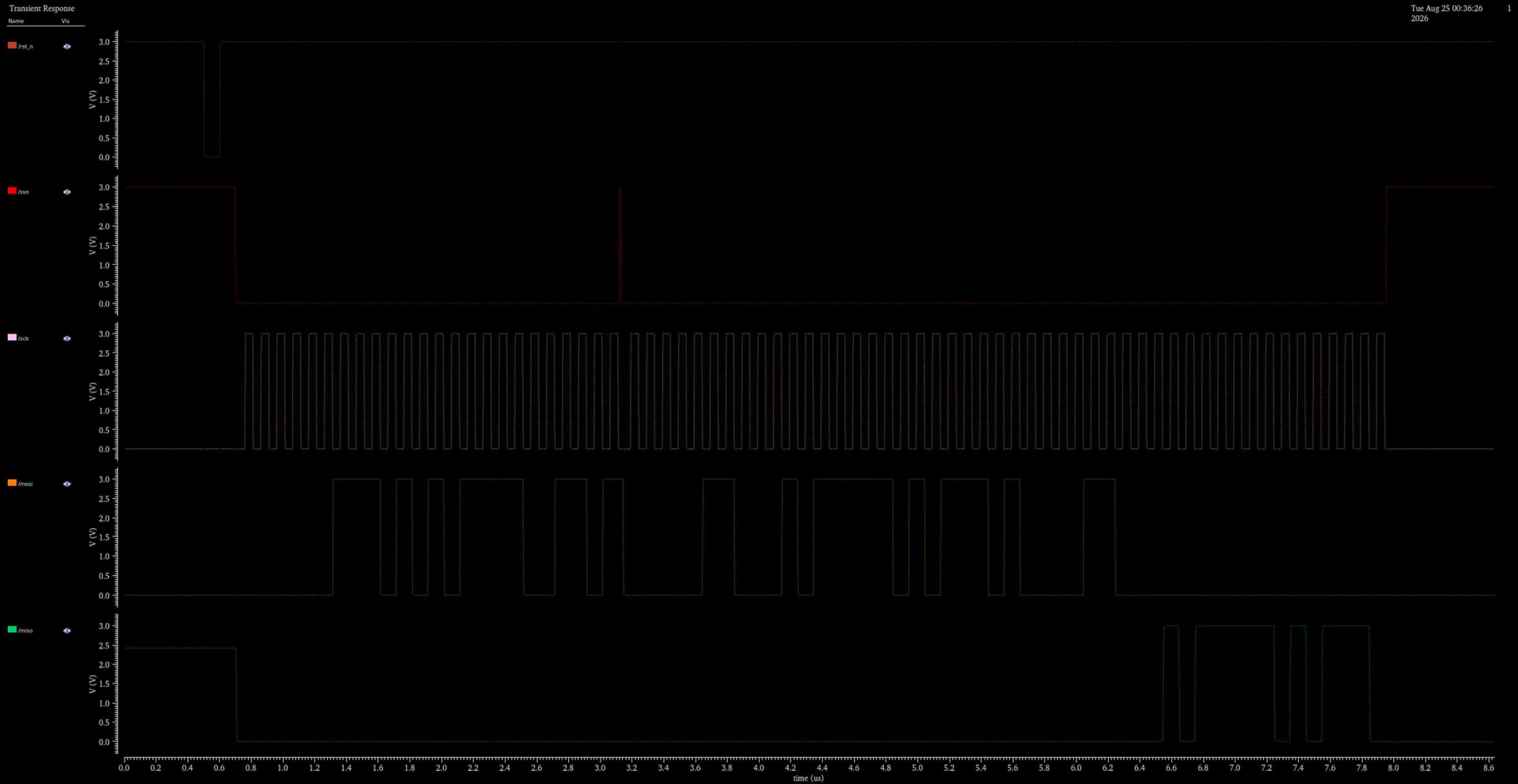


AMS Simulation

- Write an SPI Master in SystemVerilog
- Connect to SPI Slave
- Run AMS to check the transfer waveforms



AMS Simulation Passed @ ss/1.62V/3.0V/125°C, C_L=20pF

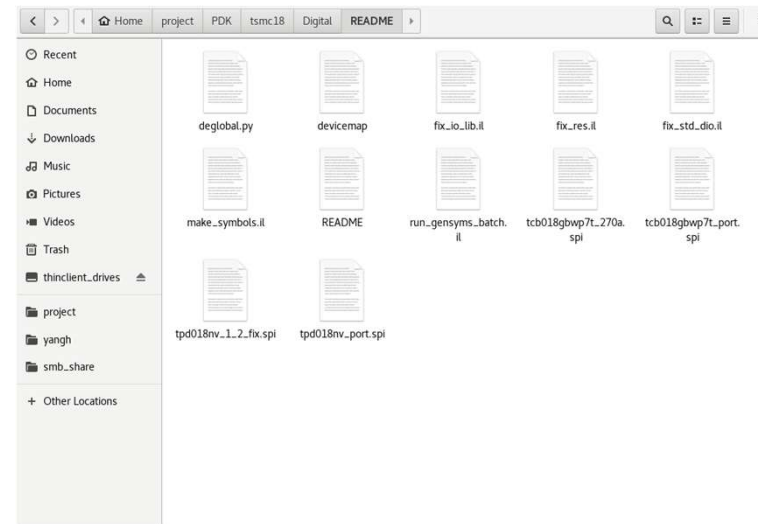
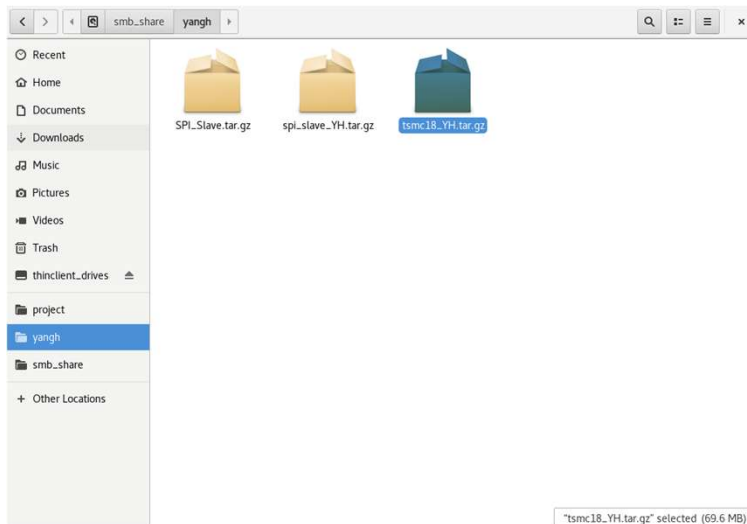


ELECTRICAL CHARACTERISTICS

PARAMETER	SYMBOL	DESCRIPTION	SPEC			UNIT
			MIN	TYP	MAX	
IO Supply	VDDPST		3.0	3.3	3.6	V
Core Supply	VDD		1.62	1.8	1.98	V
Power-on Current	I_{pk}			0.1	6	mA
Load Capacitance	C_L			20		pF
Clock Period	t_{SCK}			100		ns
ssn Setup/Hold Time	t_{SSH}/t_{SSN}	Innovus STA & testbench verified		10		ns
Minimum ssn High Time	t_{IDLE}	Innovus STA & testbench verified		10		ns
Output Enable Time	t_{EN}	ss/1.62V/3.0V/125°C, 20pF			7	ns
Output Disable Time	t_{DIS}	ss/1.62V/3.0V/125°C, 20pF			3	ns
Data Output Valid Time	t_V	ss/1.62V/3.0V/125°C, 20pF			7	ns

MISC

- Resistor/Diode mismatches in IO library have all been fixed, all IOs in the library are now available
- All scripts are included in tsmc18/Digital/README
- Check my smb_share to use the updated version



Thanks!

